

Assignment # 1

1. Design a web page using all tags which are practiced in lab.
2. Use Text Format tags, anchor tag, list tags, image tag, comment tag where needed.
3. Use decent color scheme, use images to beautify it. Underline shows hyperlink.

Start from here 

Internet & Web Fundamentals

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|---|---|
| <u>1. The Internet</u> | <u>13. URI — Uniform Resource Identifier</u> |
| <u>2. The World Wide Web (Web)</u> | <u>14. HTTP — HyperText Transfer Protocol</u> |
| <u>3. W3C — World Wide Web Consortium</u> | <u>15. HTTPS — HyperText Transfer Protocol Secure</u> |
| <u>4. Web Page</u> | <u>16. SSL — Secure Sockets Layer</u> |
| <u>5. Website</u> | <u>17. TLS — Transport Layer Security</u> |
| <u>6. Home Page</u> | <u>18. HTML — HyperText Markup Language</u> |
| <u>7. Index Page</u> | <u>19. IP Address</u> |
| <u>8. Landing Page</u> | <u>20. DNS — Domain Name System</u> |
| <u>9. Web Browser</u> | <u>21. NSlookup / nslookup</u> |
| <u>10. Web Server</u> | <u>22. Search Engines</u> |
| <u>11. Web Domain</u> | <u>23. ChatGPT</u> |
| <u>12. URL — Uniform Resource Locator</u> | <u>24. Hyperlinks / Links</u> |

1. The Internet

Internet is a global network of interconnected computer networks that communicate using standard protocols such as TCP/IP.

- **ARPANET (1969):** One of the earliest operational packet-switching networks and an important predecessor of today's Internet.
- The Internet connects computers, servers, smartphones, networks, and other devices worldwide.

2. The World Wide Web (Web)

World Wide Web (WWW) is a collection of digital or electronic resources available via Internet. Web is a **service** that operates over the Internet and allows users to access interconnected web pages through hyperlinks.

- Invented by Tim Berners-Lee in 1989.
- The Web uses technologies such as HTML, HTTP, URLs, and web browsers.

3. W3C — World Wide Web Consortium

The W3C develops and maintains standards and recommendations for the Web.

Examples include standards related to:

- HTML
 - CSS
 - Web accessibility
 - Web architecture
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4. Web Page

A web page is a single document available on the Web, usually created using HTML.

Example:

<https://example.com/about.html>

The About page is one web page.

5. Website

A website is a collection of related web pages and resources available under a common domain.

Example:

<https://www.example.com>

A website may contain:

- Home page
 - About page
 - Contact page
 - Services page
 - Blog pages
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6. Home Page

The home page is generally the main or introductory page of a website.

Example:

<https://www.example.com/>

It normally provides navigation to other important sections.

7. Index Page

An index page is commonly the default page served when a directory or website location is accessed.

Typical filenames include:

index.html

index.php

index.htm

For example:

<https://example.com/>

may automatically load:

index.html

Important: A home page and an index page are related concepts but are not necessarily identical.

8. Landing Page

A landing page is a web page designed for a specific purpose, often after a user clicks an advertisement, email link, or search result.

Typical goals:

- Register for a course
 - Buy a product
 - Download an application
 - Submit a form
-

9. Web Browser

A web browser is software used to access and display web pages.

Examples:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

A browser sends requests to web servers and renders the returned content.

10. Web Server

A web server is a computer/system and software that stores, processes, and delivers web resources to clients.

Common web-server software includes:

- Apache
- Nginx
- Microsoft IIS

Basic communication:

Web Browser — HTTP/HTTPS Request —> Web Server

Web Browser <— HTTP/HTTPS Response — Web Server

11. Web Domain

A domain name is a human-readable name used to identify a website or Internet service.

Example:

www.example.com

Instead of remembering an IP address such as:

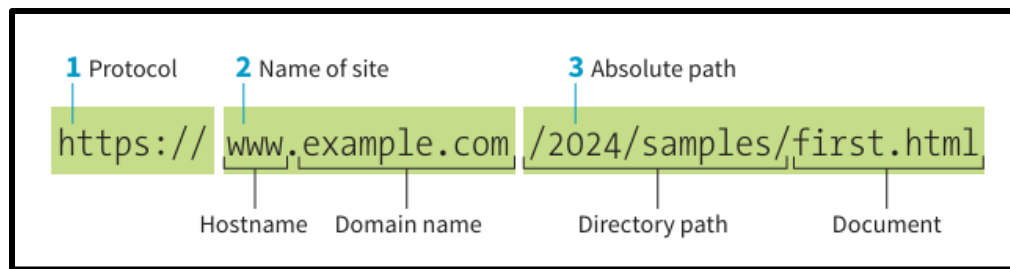
93.184.216.34

users can use the domain name.

12. URL — Uniform Resource Locator

A URL specifies the location of a resource and how to access it.

Example: (image)



13. URI — Uniform Resource Identifier

A URI is a broader concept used to identify a resource.

A URL is a type of URI.

Simple distinction:

URI identifies a resource; URL identifies a resource and provides its location/access method.

14. HTTP — HyperText Transfer Protocol

HTTP is the protocol used for communication between web clients and servers.

Example:

Browser → HTTP Request → Server

Browser ← HTTP Response ← Server

HTTP commonly uses **port 80**.

15. HTTPS — HyperText Transfer Protocol Secure

HTTPS is HTTP communication protected using modern cryptographic security, primarily through TLS.

It provides:

- Encryption
- Authentication
- Data integrity

HTTPS commonly uses **port 443**.

16. SSL — Secure Sockets Layer

SSL was an earlier security protocol designed to secure network communication.

Important: SSL has been **deprecated** and replaced by TLS. Modern HTTPS uses TLS, not SSL.

17. TLS — Transport Layer Security

TLS is the modern cryptographic protocol used to secure communications such as HTTPS.

A simplified model:

18. HTML — HyperText Markup Language

HTML is the standard markup language **used to structure** web pages.

Basic Structure of a web page also called boilerplate.

```
<!DOCTYPE html>
```

```
<html>
<head>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta charset="utf-8">
    <title>Web Page</title>
</head>
<body>
    ---
</body>
</html>
```

19. IP Address

An IP address is a unique numeric address that identifies a device on the internet or a local network.

Example IPv4: 192.168.1.10

Example IPv6: 2001:db8::1

A domain name is translated into an IP address so that network communication can take place.

20. DNS — Domain Name System

DNS translates human-readable domain names into IP addresses and performs other naming functions.

Example:

www.example.com → DNS → 93.184.216.34

21. NSlookup / nslookup

nslookup is a command-line tool used to query DNS information.

Example:

nslookup example.com

It can help determine:

- IP addresses associated with domains
 - DNS server information
 - DNS records
-

22. Search Engines

A search engine is a service that helps users find information and web pages on the Internet.

Examples:

- Google
- Bing
- DuckDuckGo

Basic process:

User enters query → Search Engine → Search Index → Relevant Results → User selects a result

23. ChatGPT

ChatGPT is an AI-based conversational system developed by OpenAI. It can understand natural-language prompts and generate responses, explanations, summaries, code, and other content.

Important distinction:

A search engine primarily helps users find existing information, while ChatGPT primarily generates responses based on its model and available information/tools.

24. Hyperlinks / Links

A hyperlink is a clickable reference that connects one resource to another.

HTML example:

```
<a href="https://example.com">Visit Example</a>
```

Links can connect:

- One web page to another
 - One section to another
 - A website to another website
 - A page to a file or resource
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